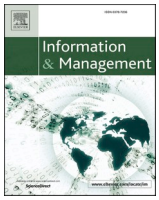
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How social crowding impacts mobile shopping: A perspective from information processing

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ABSTRACT

This research investigates how social crowding affects mobile purchase behavior through a scenario-based questionnaire, field research, and eye-tracking experiment, from the perspective of information processing. We find that social crowding reduces consumers' purchase intentions, mediated by focused attention. This impact is more pronounced for search products compared to experience products, highlighting the moderating role of product type. Eye-tracking results show that under high social crowding, total fixation counts decrease only for search products and mediate the effect of social crowding on mobile consumption, supporting the idea that reduced focused attention impedes information processing and lowers purchase intentions across product types.

1. Introduction

As reported by Redline Digital, worldwide mobile e-commerce sales have reached $2.2 trillion in 2023, making up 60 % of all e-commerce sales globally [61]. Mobile e-commerce has increasingly integrated into our daily lives, allowing consumers to shop anytime and anywhere using their mobile devices [83,106,123]. Consumers can engage in mobile shopping in any environment, such as crowded subway stations, noisy restaurants, or while traveling to popular tourist destinations [1]. In these shopping environments, consumers inevitably encounter crowded scenarios, such as during peak hours in public transit systems, waiting in line at popular stores, or at large-scale events like concerts or sports games. Much research has shown that social crowding can significantly impact consumer behavior, including affective reaction [27], shopping satisfaction [14,108], and decision-making [76]. For instance, Puzakova & Kwak (2017) suggested that consumers' perceptions of an anthropomorphized brand's intention to interact with them in crowded environments trigger greater social withdrawal, leading to lower brand preference [94].

Given the significance of mobile shopping, researchers have attempted to examine various factors that can influence consumer behavior in mobile shopping, such as interface design [25,134], user trust [40], endorsers [59], and personalized content [47]. However, limited literature on mobile shopping has paid attention to environmental factors, such as the social crowding environment. While the effects of social crowding have been widely examined in offline retail contexts, mobile shopping presents a different purchasing context in which consumers access and process product information in fundamentally different ways [22,96]. In offline retail shopping, consumers can interact directly with physical products, observe real-time social cues, and rely on multisensory input to aid their evaluations [22,98]. In contrast, mobile shopping limits consumers to a small screen interface, where product information is primarily text- and image-based, often requiring more deliberate cognitive resources to extract, compare, and evaluate key details [41,68]. Previous research has highlighted the critical role of information processing in shaping consumer purchase behavior [42,120]. Given the differences in information processing between offline and mobile shopping, the influence of social crowding may not operate in the same way across contexts. Therefore, this study focuses on the mobile shopping context, investigating how social crowding influences consumer behavior through the perspective of information processing, aiming to provide actionable insights for optimizing mobile commerce strategies.

In mobile shopping, products are typically categorized as search and

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0378-7206/© 2025 Elsevier B.V. All rights are reserved, including those for text and data mining, AI training, and similar technologies.

experience products [131]. Extensive literature has revealed consumers' distinct patterns in behaviors and information processing toward these two types of products [52]. To get more in-depth research results, we further explore whether and how the product type interacts with social crowding to affect consumer mobile consumption behaviors. This is essential for retailers to develop more effective strategies for mobile marketing that offer pathways for optimizing product recommendation systems and webpage design in mobile e-commerce.

To address our research questions, we apply the environmental load theory, which suggests that when environmental stimuli, such as social crowding, exceed consumers' processing threshold, they may selectively ignore certain information or simplify decision-making to manage the overload [107]. This adjustment may influence consumers' focused attention in information processing [66] and subsequent consumer behavior [65]. Additionally, previous studies suggest that consumers exhibit different information processing patterns depending on the product type and environmental factors. For instance, Doucé & Adams (2020) reported that consumers' information processing is disrupted by sensory overload (e.g., light, scent, music), where excessive stimulation hinders attention span and processing fluency, impacting product evaluations [27]. However, information processing is a continuous and dynamic process [116]. To capture this process, we use eye-tracking to monitor consumers' visual attention while browsing product pages on mobile devices. This method allows us to assess cognitive load related to various product information and gain insights into information processing through attention distribution analysis [3,33]. In doing so, we provide a deeper understanding of the mechanisms influencing mobile consumption behavior by exploring how the product type moderates the impact of social crowding based on the environmental load theory.

2. Literature review

2.1. Social crowding and mobile shopping

Social crowding refers to the subjective feeling and evaluation of the environment [26,44,53], often measured in terms of perceived crowding [30,109,118]. It has been confirmed to influence consumer product evaluation [43,94,128] and behavior [5,37,100,112]. The existing literature on the effect of social crowding on consumer behavior is mixed. On the one hand, previous research has found that a crowded retail environment interacts with consumers' ability to evaluate product information, thereby reducing their purchasing confidence [93]. In response, consumers tend to adopt some debugging behaviors, such as minimizing their shopping time, reducing exploratory behavior, or even leaving the store without purchasing anything [5,93]. Maeng & Tanner (2013) explained that social crowding activates avoidance motivation, leading to a stronger prevention focus and reducing consumers' risk appetite. Furthermore, in crowded environments, some literature suggests that consumers also tend to rely on concrete low-level construal [75]. Coskun et al. (2019) further explained that this occurs because consumers are more likely to feel nervous and confused in a crowded environment [20]. On the other hand, research has also identified positive effects of social crowding on experiential consumption, showing that social crowding can promote the reputation of experiential consumption and compensatory consumption [18,69]. For example, Gong et al. (2023) illustrated that social crowding motivates consumers to engage in minimalistic consumption as a means of coping with their experience of chaos and maintaining their psychological need for order [37]. In offline settings, social crowding affects consumers through physical space constraints, movement restrictions, and direct social interactions.

With the development of mobile e-commerce, consumers can engage in mobile consumption in various locations [67], broadening the research scope and significance of social crowding in this context. However, given the differences in information processing between offline and mobile shopping, findings from offline retail crowding may not directly apply to mobile contexts. Offline shopping involves direct product interaction and multisensory input [22,98], while mobile shopping relies on text- and image-based information on small screens, requiring more focused cognitive resources [41,68]. These differences suggest that social crowding may influence decision-making through distinct mechanisms in mobile contexts [1,12]. Emerging studies have begun exploring mobile shopping under social crowding. For example, Andrews et al. (2016) found that commuters in crowded subway trains were twice as likely to respond to a mobile offer compared to those in non-crowded settings. They attributed this to mobile immersion, suggesting that when physical space is invaded, individuals withdraw inward and become more receptive to mobile advertising [1]. Extending this line of inquiry, Consiglio et al. (2018) demonstrated that social crowding can also promote interactive behaviors, such as increased willingness to engage in word-of-mouth sharing in mobile environments [18]. However, this line of research is still in its infancy, and the phenomenon and mechanisms, such as the direct effect of social crowding on mobile purchase intentions, remain unclear. Moreover, we summarize the main findings of recent literature on social crowding and highlight their limitations in relation to the present research in Table 1.

2.2. Social crowding, focused attention, and information processing

According to the environmental load theory, cues in the environment can lead to cognitive overload [27,102], which can lower focused attention among individuals [48]. This overload implies that individuals' information processing can be significantly affected by complex environments [9,82], wherein focused attention plays a crucial role in navigating through environmental stimuli [38,65]. Focused attention refers to the subjective intention to direct and sustain cognitive resources on a specific target while minimizing distractions [23]. It reflects an individual's conscious effort to remain engaged with relevant information or tasks, often assessed through self-report [88,124]. The theory of information processing suggests that to effectively encode information, individuals typically first process sensory information from the environment and then selectively pay attention to the target information while ignoring distractions in that environment [80]. This process is strictly constrained by the limited capacity of human attention [28]. While individuals are exposed to vast amounts of information, only a small portion can be deeply processed, making focused attention a critical bottleneck in information processing [85,121,126]. As focused attention declines, individuals struggle to maintain focus [6], affecting both the quantity and quality of attention [31]. This limitation influences how information is processed, subsequently impacting consumer behavior [72,129].

Social crowding environments present a unique challenge by inundating consumers with excessive stimuli that can overwhelm their information processing capacities [29,37]. Milgram (1970) suggested that one adaptive response to such overload is to reduce the time and attention allocated to each input, which can distort behavior [82]. Hock & Bagchi (2018) further supported this by showing that exposure to a highly stimulating social crowding environment distracts consumers' attention, which affects their information processing patterns and ultimately influences their decision-making [48]. Thus, grounded in the environmental load theory, this research will focus on how social crowding affects mobile shopping behavior from the perspective of information processing.

2.3. Product type

In mobile shopping, it was reported that product type influences consumers' evaluation of the product, leading to different information processing patterns [13,34,78,125,134]. For example, Rhee and Choi (2020) appraised that compared with the experience products, the search products need higher involvement and attention in information processing [97]. For search products, individuals evaluate them based

Table 1
Summary of recent literature on social crowding.

Author (Year) & Journal	Research background	Main research findings	Limitations related to current study
Gong et al. (2023) [37] Journal of Business Research	Offline retail environment	Social crowding can prompt consumers to adopt minimalist consumption behaviors, such as reducing the number of possessions and preferring minimalist designs. However, this effect diminishes when the crowd includes familiar individuals or when consumers have a more interdependent self-construal.	It explores how psychological order needs drive offline minimalist consumption under social crowding, moderated by group composition. However, this effect may not extend to mobile shopping, where social presence is less immediate and group influence is reduced.
Liao et al. (2023) [69] Computers and Industrial Engineering	Offline retail environment	Social crowding increases consumers' willingness to buy green products, partially mediated by face consciousness (gaining or losing face). Price levels moderate this effect—higher-priced green products are preferred in high-crowding contexts, while lower-priced ones are favored in low-crowding settings.	It examines the mediating role of face consciousness between social crowding and consumers' willingness to purchase green aquatic products. However, face consciousness is less applicable to mobile shopping, which involves more autonomous and independent decision-making.
Gupta & Coskun (2021) [43] Journal of Retailing and Consumer Services	Offline retail environment	Social crowding reduces purchase intentions, especially for edible goods, by increasing perceived contamination. This effect is weaker for non-edible products. However, perceived scarcity mitigates the negative impact of crowding, boosting purchase intentions for edible goods.	It examines the impact of crowding and store clutter on consumers' willingness to purchase in offline retail. However, store clutter is not a factor in mobile shopping.
Coskun et al. (2019) [20] Journal of Consumer Behaviour	Offline retail environment	Crowding and store disarray increase shopping confusion and reduce purchase intentions. Perceived contamination mediates this effect, while scarcity perception can help offset the negative impact.	It primarily examines the impact of crowding and store clutter on shoppers' confusion and behavioral intentions. However, store clutter is not a factor in mobile shopping.
Consiglio et al. (2018) [18] Journal of Consumer Research	Offline retail environment	The threat of social crowding to consumers' sense of control motivates them to spread more word-of-mouth information to regain control.	It primarily focuses on how the threat to consumers' sense of control caused by social crowding influences their word-of-mouth communication
		Especially in crowded service scenarios, consumers are inclined to regain a sense of control by leaving reviews about products and services.	about products and services, rather than directly examining consumer behavior.
Hock & Bagchi (2018) [48] Journal of Consumer Research	Restaurant environment	Social crowding increases calorie consumption by prompting choices of high-calorie foods and greater intake. Crowd-induced distraction impairs cognitive processing, leading consumers to rely more on emotions than reasoning, thus favoring indulgent food options.	It examines the impact of crowded environments on consumers' calorie consumption from the perspective of cognitive processing ability but does not consider boundary variables.
Huang et al. (2018) [53] Journal of Consumer Research	Offline retail environment	In crowded retail environments, consumers cope by strengthening their sense of belonging, often through brands. Under high crowding, reliance on brands increases, reinforcing brand attachment and encouraging repeat purchases.	It proposes that social crowding increases brand attachment by triggering social avoidance. However, this effect may differ in mobile shopping, which lacks face-to-face interaction.
Wilson & Suh (2018) [122] International Journal of Advertising	Offline advertising	As crowd density increases, consumers' attention to place-based advertisements initially declines and then rises, forming a U-shaped pattern. At high density, social proof effects may emerge, as individuals are drawn to ads by observing others' gaze and behavior.	It focuses on the impact of social crowding on consumers' attention to advertising in offline settings. However, the way advertising content is presented differs significantly from how product information is displayed in mobile shopping environments.

on objective attributes [95], which typically involves engaging in detailed comparisons and analysis of various information [55,72,77]. This process involves organizing, analyzing, and weighing information, which demands more focused attention on specific product parameters [60,87,97].

On the other hand, for experience products, consumers evaluate them heavily on their subjective assessments [52,58,72,110], which may involve intuition, personal preferences, emotional tendencies, or simply general impressions for judgment [8,72,77]. It involves a more holistic and less detailed approach to information processing. According to previous studies, purchase decisions are made following the application of attention to process product information and subsequent evaluation during information processing [7,19,35,57]. Thus, we also consider product type as an essential variable, as it fundamentally shapes how consumers allocate their attention and information processing in social crowding.

3. Research model and hypothesis development

Fig. 1 illustrates our research model. Grounded in the environmental load theory and the perspective of information processing, this research examines how social crowding influences consumers' mobile purchase intentions, with focused attention as a potential mediator. In addition, the moderating role of product type is also explored.

3.1. Social crowding and mobile purchase intentions

As discussed in the literature review part, social crowding can influence consumer behavior in different scenarios. According to the environmental load theory, individuals in crowded environments are exposed to excessive environmental stimuli, triggering adaptive behavioral responses [27,102]. In the context of social crowding, consumers engaging in mobile shopping encounter dual stimuli: external disturbances or sensory stimuli caused by the environment and their cognitive demand for processing product information on the mobile interface. Therefore, consumers need to process diverse stimuli. These leads consume individuals more cognitive resources [10], reducing their ability to process information [48,81]. In mobile shopping, consumers primarily rely on processing online information to evaluate products before making purchase decisions [32]. When their ability to process the information is compromised, it is difficult to effectively process important, relevant information, which can lead to inadequate evaluation of the product. Consumers are more likely to form inaccurate perceptions of product value or availability [84,92]. As a result, it can lead to hesitation or avoidance in decision-making [75], ultimately decreasing their purchase intentions [54]. Consequently, we hypothesize that in a high social crowding environment, consumers tend to lower their purchase intentions. Based on the above, we suggest the following hypothesis:

H1. Social crowding has a negative impact on consumers' mobile purchase intentions.

3.2. The mediating effect of focused attention

In our research context, focused attention denotes the degree to which individuals consciously direct their attention to the product present on the mobile device. According to the environmental load theory, social crowding induces a perceptual state in which individuals are exposed to an excessive volume of external stimuli, such as proximity of others and movement of others within a limited space [73,79,109]. These multiple stimuli compete for individuals' attentional resources and occupy their cognitive recourse [26,27,102]. This increased input taxes the attentional system, reducing individuals' ability to maintain focused attention on the focal task [29,82]. In high social crowding environments, the abundance of external stimuli and social presence intensifies this effect, significantly depleting available cognitive resources and disrupting consumers' focused attention [48,81]. In contrast, low social crowding environments involve fewer external distractions, allowing individuals to maintain a higher level of attentional focus on product-related information [48]. The decline in focused attention impedes information processing, making it more difficult for consumers to thoughtfully evaluate and analyze product information [48], which will, in turn, lead to decreased purchase intentions. Therefore, focused attention serves as a mediating mechanism through which social crowding influences consumers' mobile purchase intentions. Since it was, we propose:

H2. Focused attention mediates the effect of social crowding on consumers' mobile purchase intentions.

3.3. The moderating effect of product type

As mentioned previously, mobile shopping typically involves two types of products: search products and experience products [87,110]. For search products, consumers' purchase decisions mainly rely on detailed parameter analysis and functional comparisons, requiring rational evaluation [127]. To make informed consumption decisions, consumers require a high level of focused attention to effectively acquire and process key information [17,56]. When individuals need to process difficult or unfamiliar information, such as product parameters, they should mobilize greater cognitive resources and sustain a higher level of attention to effectively engage with the material [64,101,119]. However, in high social crowding environments, external stimuli increase and compete for consumers' cognitive resources, making it difficult to maintain a high level of focused attention [48]. Thereby, we propose:

H3a. For search products, consumers' focused attention in high social crowding will be significantly lower than that in low social crowding.

Focused attention plays a critical role in determining how attention is allocated during information processing [80]. It affects not only where consumers direct their attention but also how deeply they process the information [62]. When consumers' focused attention is weakened by high social crowding, their attention tends to prefer simplified, easily processed information and avoid cognitively demanding content, such as detailed product parameter information [85,121,126]. This behavioral tendency can lead to skipping important details and reducing in-depth reading and comparison of key product parameters [17,36,56,111]. This hinders sufficient information processing for search products. Consequently, the decline in focused attention directly impairs information processing, making it more difficult for consumers to fully understand the product and lower their purchase intentions. Thereby, we propose:

H4a. For search products, consumers' mobile purchase intentions in high social crowding will be significantly lower than those in low social crowding.

In contrast, for experience products, consumers' purchase decisions mainly rely on intuitive judgment, visual stimuli, and user reviews, such as images and reviews [8,51], rather than on analytical processing of product parameters [72]. This type of information is generally less complex, more heuristic-based, and easier to process, requiring fewer cognitive resources [64]. Because cognitive demand is low, consumers'

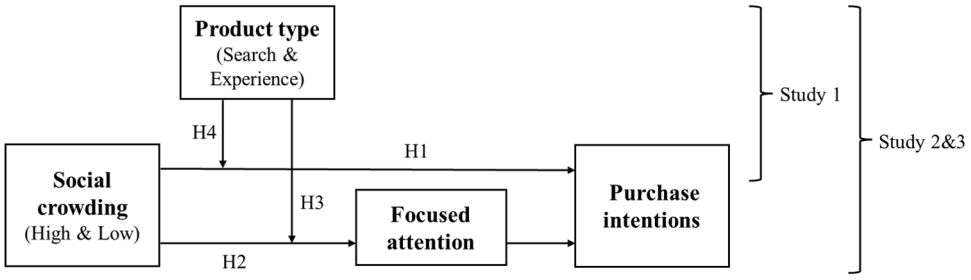


Fig. 1. Research model.

focused attention is less likely to be affected by environmental distractions [62]. Therefore, we consider that the focused attention directed toward experience products remains relatively stable, even under high social crowding conditions. As focused attention remains stable, information processing is unaffected, resulting in no significant change in consumers' mobile purchase intentions for experience products. Therefore, we propose:

H3b. Compared to search products, the difference in consumers' focused attention between high and low social crowding will be less pronounced for experience products.

H4b. Compared to search products, the difference in consumers' mobile purchase intentions between high and low social crowding will be less pronounced for experience products.

4. Overview of studies

We conducted three studies to test our hypotheses, as detailed in Sections 5, 6, and 7. Study 1 used a scenario-based questionnaire to explore the negative impact of social crowding on consumers' mobile purchase intentions and the moderating effect of the product type. Study 2 conducted field research to reverify findings of Study 1 and further examine the mediating role of focused attention across different product types. Study 3 employed an eye-tracking experiment in ecologically valid settings (social crowding in the laboratory) to explore how social crowding affects mobile purchase intentions by influencing information processing. The conceptual model is depicted in Fig. 1.

5. Study 1

This study aims to explore the effect of social crowding on consumers' purchase intentions and examine the moderating effect of product type. We conducted a scenario-based questionnaire with 2 social crowding (high vs. low) × 2 product type (search vs. experience) design.

5.1. Pretest

5.1.1. Manipulation of social crowding

According to previous studies, the perception of social crowding can be induced by picture priming [76]. Thus, we employed picture priming to manipulate social crowding in this scenario-based questionnaire. Based on the definition of social crowding [109], 10 pictures depicting high social crowding and 10 pictures depicting low social crowding were selected from the internet. Then, 33 participants (15 males, $M_{age} = 22$, $SD = 2.41$) were asked to evaluate the perceived crowding of these 20 candidate images on a 7-point scale using 5 items (as detailed in Appendix A) adopted from Machleit [74]. The picture with the highest perceived crowding score was selected as the priming picture for the high social crowding group, while the picture with the lowest score was selected for the low social crowding group ($M_{high} = 6.52$, $SD = 0.86$; $M_{low} = 1.61$, $SD = 0.82$, t (32)=18.172, $p < 0.001$). The detailed statistical results are described in Appendix A and the two chosen pictures are shown in Appendix B.

5.1.2. Manipulation of product types

For the selection of target search and experience product, a pretest was conducted according to the following steps: first, a panel of five marketing researchers listed seven kinds of potential search products and seven kinds of potential experience products according to Klein's (1998) definition of search (i.e., USB drive) and experience products (i.e., sunglasses) [60]. Then, 40 participants were asked to evaluate the search and experience characteristics of the 14 candidate products by using four items on a 5-point scale measure, which were adapted from Senecal and Nantel (2004) [103]. The four items assessed whether participants could obtain and evaluate product information: (1) before purchase, (2) mostly before purchase, (3) mostly after purchase, and (4) only after purchase. Details are presented in Appendix A. Items (1) and (2) measured search indicators, while items (3) and (4) measured experience indicators. We selected three products with the highest search index (e.g., weight scale, USB drive, power strip) and three with the highest experience index (e.g., electric toothbrush, sunglasses, Bluetooth headset) as the experimental materials. The statistical results are shown in Appendix A. In Studies 1 and 2, USB drives were used as search products and sunglasses as experience products. In Study 3, weight scales, USB drives, and power strips were chosen as search products, while electric toothbrushes, sunglasses, and Bluetooth headsets were selected as experience products.

To standardize the stimuli, images of products priced between 95 CNY to 105 CNY (averaging 100 CNY) were sourced from a mobile shopping platform. The images were edited by Photoshop 13.0 to produce a uniform format, white background, etc. (Adobe Systems Incorporated, San Jose, CA, USA). Pictures of the six products are shown in Appendix B.

5.2. Participants

We recruited 437 participants through an online questionnaire platform, with each taking 1–2 minutes to complete the questionnaire and receiving 3 CNY for compensation. After excluding 6 incomplete responses, we had a total of 431 valid participants (156 males, aged 18–56, $M_{age} = 26.91$, $SD = 8.98$). Participants were randomly divided into four groups ($N_{high\text{-}search} = 117$, $M_{age} = 27.72$, $SD = 9.26$; $N_{high\text{-}experience} = 113$, $M_{age} = 25.67$, $SD = 8.23$; $N_{low\text{-}search} = 82$, $M_{age} = 25.67$, $SD = 9.09$; $N_{low\text{-}experience} = 119$, $M_{age} = 27.85$, $SD = 7.67$). All participants had over two years of mobile shopping experience, spent an average of 13.63 ± 13.12 h shopping on mobile devices in the past three months, and had an average of 5.07 ± 3.04 years of mobile shopping experience. The experiment received approval from the university's Internal Review Board (IRB Number: 2021BC012).

5.3. Research setting and survey

Participants were first shown a selected image depicting either high or low social crowding. Then, they were asked to imagine themselves in the depicted crowding scenario while attempting to purchase sunglasses or USB drives through *Taobao*, a popular mobile shopping app in China, which participants were familiar with. After reviewing the product image, participants reported their purchase intentions using three items on a 7-point Likert scale (Cronbach's α = 0.94), adapted from Chiu et al. (2012) [16], as detailed in Appendix A. The perceived crowding was also measured using the 5 items (Cronbach's α = 0.93) described in the pretest as a manipulation check.

5.4. Data analysis and results

5.4.1. Manipulation checks of social crowding

We averaged the scores of five items to reflect the perceived crowding and conducted an independent-samples *t*-test to compare perceived crowding in high and low social crowding conditions. The results showed that perceived crowding in the high social crowding conditions was significantly higher than in the low social crowding conditions ($M_{high} = 5.65$, SD = 1.05; $M_{low} = 2.41$, SD = 1.27; t (429)=−28.729, $p < 0.001$).

5.4.2. Effect of social crowding on purchase intentions for search and experience products

Univariate analysis of variance revealed significant main effects of social crowding (F (1, 427)=164.287, $p < 0.001$, $\eta_p^2 = 0.278$) and product type (F (1, 427)=63.201, $p < 0.001$, $\eta_p^2 = 0.129$) on purchase intentions. Overall, purchase intentions were lower in high social

crowding (M_{high} = 3.39, SE = 0.08) compared to low social crowding (M_{low} = 4.85, SE = 0.08), supporting H1. Purchase intentions were lower for the search product (M_{search} = 3.67, SE = 0.08) than for the experience product ($M_{experience}$ = 4.58, SE = 0.08). Meanwhile, the interaction effect between social crowding and product type was also significant (F (1, 427)=136.352, $p < 0.001$, $\eta_p^2 = 0.242$). Further simple effect analysis indicated that the purchase intentions for the search product were lower in high social crowding ($M_{high\text{-}search}$ = 2.27, SE = 0.13) than in low social crowding ($M_{low\text{-}search}$ = 5.07, SE = 0.11, F (1, 427)=274.757, $p < 0.001$, $\eta_p^2 = 0.392$), supporting H3a. However, there was no significant difference for the experience product (F (1, 427)= 0.716, $p = 0.398$, $\eta_p^2 = 0.002$), supporting H3b. These results are shown in Fig. 2.

5.5. Discussion

The findings of Study 1 showed that social crowding negatively impacts consumers' mobile purchase intentions, aligning with the findings of Cai et al. (2021) [12]. We also found a moderating effect of product type, indicating that social crowding reduced purchase intentions for search products, rather than for experience products. Study 1, while informative, has left deficiencies in understanding how social crowding impacts consumers' purchase intentions. Additionally, although picture priming was effective in inducing social crowding, social crowding is a complex phenomenon linked to physical restriction, the possibility of interpersonal social interaction, and psychological pressure. Hence, we intend to conduct field research to further explore these issues.

6. Study 2

Through field research, we aim to explore the moderating effect of product type and the mediating effect of focused attention on the impact of social crowding on mobile purchase intention in authentic social crowding environments. We applied a 2 social crowding (high vs. low: between-subject) × 2 product type (search vs. experience: within-subject) mixed design.

6.1. Participants

Two hundred subway passengers (108 males, aged 20–50, M_{age} = 23.95, $SD = 5.17$) were randomly selected to participate in this research. Each participant received an additional fee of 5 CNY. Participants were divided into two groups: the high social crowding group (N_{high} = 100, aged 20–50, M_{age} = 23.45, SD = 4.53) and the low social crowding group (N_{low} = 100, aged 20–50, M_{age} = 24.45, SD = 5.72). On average, participants took the subway or bus 4.48 ± 4.16 times per week. All participants had prior experience of using mobile phones while on the subway or bus, with 77.5 % indicating that they had engaged in mobile shopping during these commutes. Additionally, 94 % of participants have over 2 years of mobile shopping experience, and 56 % shopped >10 times on mobile terminals in the past three months. The experiment received approval from the university's Internal Review Board (IRB Number: 2024BC015).

6.2. Experiment protocol

Social crowding. Based on Shanghai Metro Line 9, we collected data at four adjacent main platforms: Xujiahui Station, Lujiabang Station, Jinqiao Road Station, and Fangdian Road Station. These stations are near dense residential areas and large passenger flows. To ensure the smooth operation of the subway, we selected passengers on board as the research subjects, excluding those who were scheduled to disembark at the next station. Passengers during peak hours were selected as participants for the high social crowding group, while passengers during the idle period of operation were selected for the low social crowding group, as shown in Appendix B.

Product type. As for the product selection, USB drives and sunglasses were selected as search and experience products, respectively, which was also selected by the pretest shown in Study 1. The pictures were edited by Photoshop 13.0 to produce a uniform format, white background, etc. (Adobe Systems Incorporated, San Jose, CA, USA).

6.3. Experiment procedure

Six graduate students (3 males and 3 females) were assigned to collect data on the metro by randomly selecting passengers for participation. Each participant typically took 1–2 minutes to complete the experiment. Due to the high rejection rate from passengers, exceeding 80 %, the data collection process was extended over two months. First, participants used their phones to scan our QR code. Then, they viewed a standardized product image that we had created for the task. The images were controlled for factors such as brand and price, consistent with Study 1. Participants were instructed that they were in the current environment and browsing for the USB drives (or sunglasses). After viewing the product images, they were asked to report their purchase intentions with a 7-point Likert scale (Cronbach's α = 0.92), adapted from Chiu et al. (2012) [16]. Furthermore, they answered four focused attention questions regarding their shopping experiences for USB drives (or sunglasses) on their phones, also using a 7-point Likert scale (Cronbach's α = 0.84) [89]. These details are presented in Appendix A.

The order of presentation for the two products was random, each occupying 50 %. Furthermore, we interspersed an unrelated distractor task (intertemporal choice task) between the two products aimed at

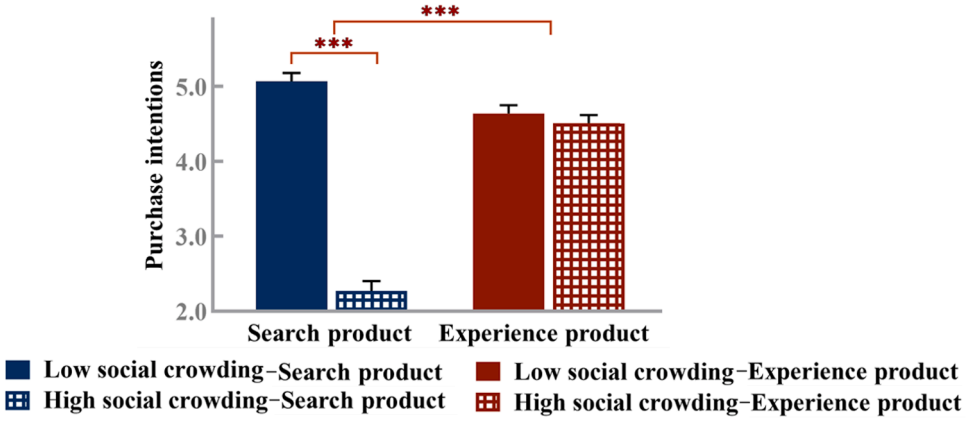


Fig. 2. Purchase intentions for search and experience products.
Error bars denote standard error across participants. ***$p < 0.001$.

mitigating participants' memory influence on the evaluation of the previous product, ensuring independent assessments for each product and controlling for experimental effects [49]. After finishing the questions about the two products, the participants needed to rate the degree of perceived crowding in the current environment with a 7-point scale (Cronbach's α = 0.89) [74], serving as the manipulation check of social crowding, utilizing the same items employed in Study 1. Finally, the participants filled out demographic information, such as gender, age, years of mobile shopping experience, and the number of mobile purchases in the past three months.

6.4. Data analysis and results

6.4.1. Manipulation checks of social crowding

The independent-samples *t*-test result showed that the perceived crowding of the participants in high social crowding was significantly higher than that in low social crowding ($M_{high} = 4.53$, $SD = 1.61$; $M_{low} = 2.37$, $SD = 1.22$; t (198)=−10.730, $p < 0.001$). The above results showed that our manipulation was robust.

6.4.2. Effect of social crowding on purchase intentions for search and experience products

A mixed design repeated measure analysis of variance (ANOVA) showed both the main effects of social crowding (F (1, 198)=9.453, $p = 0.002$, $\eta_p^2 = 0.046$) and product type (F (1, 198)=4.923, $p = 0.028$, $\eta_p^2 = 0.024$) on purchase intentions were significant. The purchase intentions were lower in high social crowding ($M_{high} = 4.00$, SE = 0.12) than in low social crowding ($M_{low} = 4.54$, SE = 0.12), H1 was supported. The purchase intentions for the search product ($M_{search} = 4.09$, SE = 0.12) were lower than the experience product ($M_{experience} = 4.44$, SE = 0.12).

Moreover, it revealed a significant interaction effect between social crowding and product type (F (1, 198)=9.259, $p = 0.003$, $\eta_p^2 = 0.045$). Further simple effect analysis indicated that the difference in purchase intentions for the search product between high ($M_{high\text{-}search} = 3.58$, SE = 0.17) and low ($M_{low\text{-}search} = 4.60$, SE = 0.17) social crowding was significant (F (1, 198)=18.783, $p < 0.001$, $\eta_p^2 = 0.087$), H4a was supported. However, there was no significant difference in purchase intentions for the experience product (F (1, 198)=0.064, $p = 0.800$, $\eta_p^2 < 0.001$), H4b was supported. The results are illustrated in Fig. 3.

6.4.3. The mediation model test with focused attention

We adopted PROCESS model 4 [45] to conduct the mediation analysis. We found that focused attention fully mediated the effect of social crowding on purchase intentions (*indirect effect* = −0.446, SE = 0.095, 95 %CI= [−0.651, −0.277]), supporting H2 (see Table 2 and Fig. 4).

Table 2
Coefficient estimates for the mediation model.

Variable	Coefficient	SE	t	*p*	*R-sq*	*F*
Outcome: FA (mediator variable)					0.066	27.956
Constant	4.520	0.095	47.771	< 0.001		
SC	−0.708	0.134	−5.287	< 0.001		
Outcome: PI (dependent variable)					0.267	72.444
Constant	1.686	0.269	6.259	< 0.001		
FA	0.630	0.055	11.461	< 0.001		
SC	−0.094	0.152	−0.619	0.536		
The indirect effect of SC on PI						
	Effect	Boot SE	95 %CI			
FA	−0.446	0.095	−0.651	−0.277		

Notes: SC = Social crowding; FA = Focused attention; PI = Purchase intentions. Bootstrap sample size = 5000.

6.4.4. The moderated mediation model test with product type as the moderating variable and focused attention as the mediating variable

We performed a moderated mediation analysis by using the PROCESS Model 7 [45], as presented in Table 3 and Fig. 5. Results showed that the index of moderated mediation was significant, showing that the indirect relationship between social crowding and purchase intentions depended on focused attention (*indirect effect* = 0.690, SE = 0.180, 95 % CI=[0.361, 1.069]) and moderated by product type ($\beta = -0.380$, SE = 0.185, 95 %CI=[−0.744, −0.016]). Specifically, the results showed that the bootstrapped conditional indirect effects were significant only on search products (*indirect effect* = −0.791, SE = 0.142, 95 %CI=[−1.098, −0.536]), but not on experience products (*indirect effect* = −0.101, SE = 0.117, 95 %CI=[−0.338, 0.126]), as indicated by the significant interaction between social crowding and product type ($\beta = 1.095$, SE = 0.262, 95 %CI=[0.580, 1.610]). Overall, these results supported H2, H3a, H3b, H4a, and H4b.

6.5. Discussion

First, the field research validated the main effect of social crowding and the moderating effect of product type, aligning with the results of Study 1. Second, we identified that focused attention played a mediating role in the impact of social crowding on consumers' mobile consumption behavior across different product types, explaining the effect of social crowding.

However, Study 2 lacked the granularity to precisely understand how focused attention is allocated and processed across different product types in social crowding, which is essential for uncovering the underlying information processing mechanisms driving consumer behavior in these settings. Consequently, we intend to conduct an eye-tracking study, offering a more direct and quantitative assessment of attentional distribution to detect information processing.

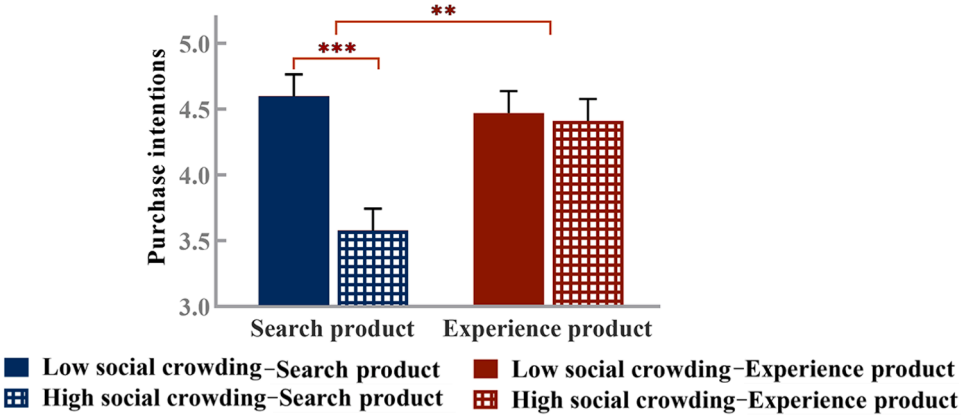


Fig. 3. Purchase intentions for search and experience products.
Notes: Error bars denote standard error across participants. **$p < 0.01$, ***$p < 0.001$.

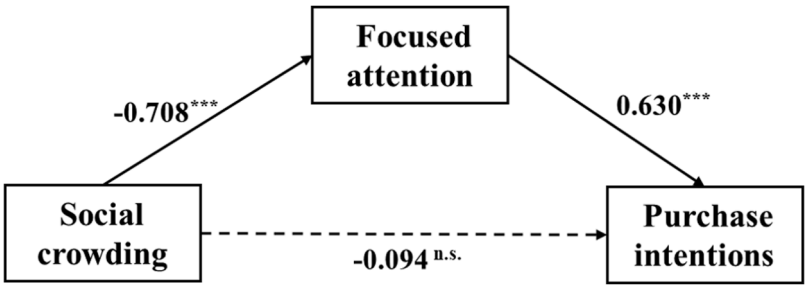


Fig. 4. Focused attention fully mediated the effect of social crowding on purchase intentions. Depicted are the standardized regression coefficients for each path of the mediation model. ***$p < 0.001$, *n.s.* - not significant.

Table 3
Coefficient estimates for the moderated mediation model.

Variable	Coefficient	SE	t	*p*	*R-sq*	*F*
Outcome: FA (mediator variable)					0.109	16.084
Constant	4.710	0.131	35.947	< 0.001		
SC	−1.255	0.185	−6.773	< 0.001		
PT	−0.380	0.185	−2.051	0.041		
SC × PT	1.095	0.262	4.179	< 0.001		
Outcome: PI (dependent variable)					0.267	2.155
Constant	1.686	0.269	6.259	< 0.001		
FA	0.630	0.055	11.461	< 0.001		
SC	−0.094	0.152	−0.619	0.536		

Notes: SC = Social crowding; FA = Focused attention; PI = Purchase intentions; PT = Product type. Bootstrap sample size = 5000.

7. Study 3

In this eye-tracking study, we intend to reveal the mechanisms of social crowding influencing consumer mobile purchase intention from the perspective of information processing. We also focused on providing direct evidence of environmental load impact on consumers' information processing and subsequent purchase intentions. Fixation counts were used to indicate attention, while the distribution of attention across AOIs was used to visually represent information processing [3,4]. We adopted a 2 social crowding (high vs. low: between-subject) × 2 product type (search vs. experience: within-subject) mixed design.

7.1. Participants

One hundred and five university students (42 males, aged 18–29, $M_{age} = 21.37$, $SD = 2.43$) were recruited and randomly assigned to the high ($N_{high} = 51$, $M_{age} = 21.61$, $SD = 2.60$) or low social crowding group ($N_{low} = 54$, $M_{age} = 21.12$, $SD = 2.22$). Each participant was paid 20 CNY. Furthermore, 96.2 % of participants had over two years of experience with mobile shopping, including using their phones to shop in social crowding. Additionally, 84.8 % had made >10 mobile shopping purchases in the past three months. All participants were native Chinese speakers, right-handed, had normal or corrected-to-normal vision, and had no history of neurological or mental illness. Written informed consent was provided by each participant before the formal experiment. The study was approved by the university's Internal Review Board (IRB Number: 2023BC133) in accordance with the 1964 Helsinki Declaration and its subsequent amendments or comparable ethical standards [2].

We only included college students as the sample in the eye-tracking study for the following reasons: first, previous studies determined that college students make up a representative sample of the e-commerce shopper population because of the high likelihood that college students are the mainstream of e-commerce and m-commerce purchases [90,130]. Second, college students were more likely to engage in the task [70], and the difference in mobile shopping in complex environments between college students and the general public may not differ substantially [90].

7.2. Experiment protocol

Materials. For the social crowding manipulation, we created a social crowding scene in the lab, drawing on methodologies from previous research [53,63], as shown in Fig. 6. Specifically, in the high social crowding condition, 7 strangers (3 males) were seated around the participant in a 10 m^2 room. In the low social crowding condition, the participant was seated in the same room with 1 stranger (random gender) and six empty chairs. To mimic a real mobile screen, we set the screen upright (21-inch with 1080 px width × 1920 px height) to display the product information pages.

For the product type, USB drives, weight scales, and power strips were selected as search products. Sunglasses, electric toothbrushes, and Bluetooth headsets were selected as experience products. These products were also selected by the pretest shown in Study 1. Each product category had five products.

The product information page was adapted according to the actual product layout on mobile devices (including a product picture, product name, transaction parameters, product reviews, and product

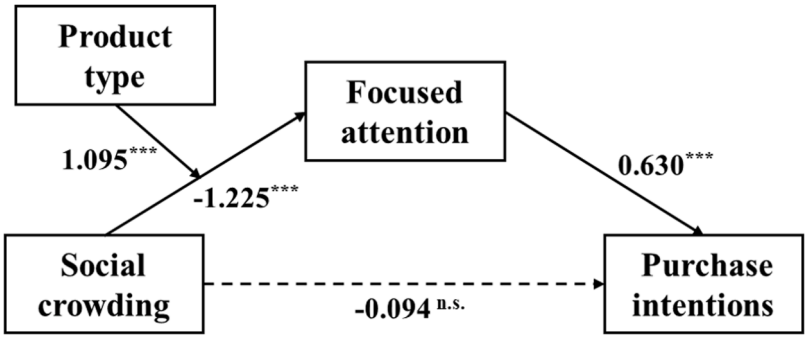


Fig. 5. Effect of social crowding on purchase intentions through moderated mediation of product type and focused attention. Depicted are the standardized regression coefficients for each path of the moderated mediation model. ***$p < 0.001$, *n.s.* - not significant.

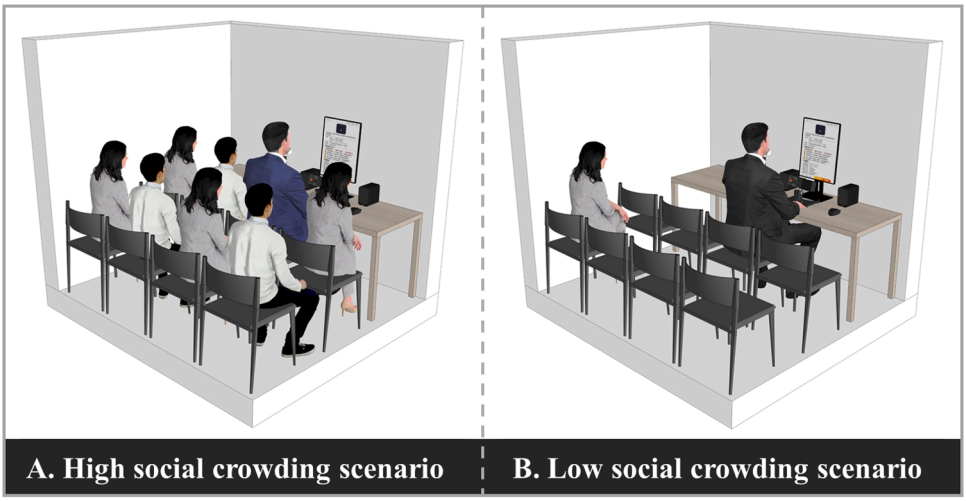


Fig. 6. Experimental schematic diagram.

parameters, as shown in Fig. 7), which all participants were familiar with. To evade the influence of price and participants' subjective preferences or brand stereotypes, the selling price and the brand name were removed [129]. The selection of product pictures was as described in Study 1. The word count for names and titles of electronic scales and toothbrushes was kept as consistent as possible. For transaction parameters, we used the same settings for both products, based on a real page from *Taobao*. The information included the place of shipment, delivery fee, product warranty, and payment method. Concerning the product reviews, we only selected the content of the positive reviews according to Zhang et al. (2014) [133], while >90 % of the reviews were indeed positive in reality. This also prevented the negative bias of review valence.

We extracted the reviews according to the following steps. First, we used LocoySpider software to crawl and extract approximately 1000 reviews for each product from *Taobao*. Next, two marketing researchers conducted an initial screening to ensure that the reviews focused on product evaluations, removing any comments related to transactions or shipping. Then, we eliminated reviews that were extremely long or short to maintain a consistent length. We further requested the researchers to check and sift the reviews, leaving 25 distinct reviews for each product category, meaning 5 different reviews per product. For the product parameters, by browsing and summarizing the parameter information of 30 similar products on *Taobao*, we selected the 7 most frequent parameter information of each product (i.e., electronic scale: product size, boot mode, and so on).

Procedure. Before the experiment began, participants were briefed on the equipment and procedures and signed an informed consent form. They then read the instructions and completed three practice trials using materials different from those used in the formal experiment. After the experimenter calibrated and validated the eye tracker, participants proceeded to the formal experiment. The experiment took place in a sound-insulated laboratory with adjustable lighting.

The participants were told to imagine they intended to buy an electric toothbrush or any other product in the current environment, through *Taobao*. Their task was to report purchase intentions (using ratings of 1–7) for the shown product by modulating the progress bar using a compatible keypad attached to their hands.

The experiment consisted of 30 trials. In each trial, participants were shown a fixation cross (400–600 ms), followed by a blank page (600 ms), and then a product information page (timeless), sequentially. When participants felt they had browsed the information sufficiently, they

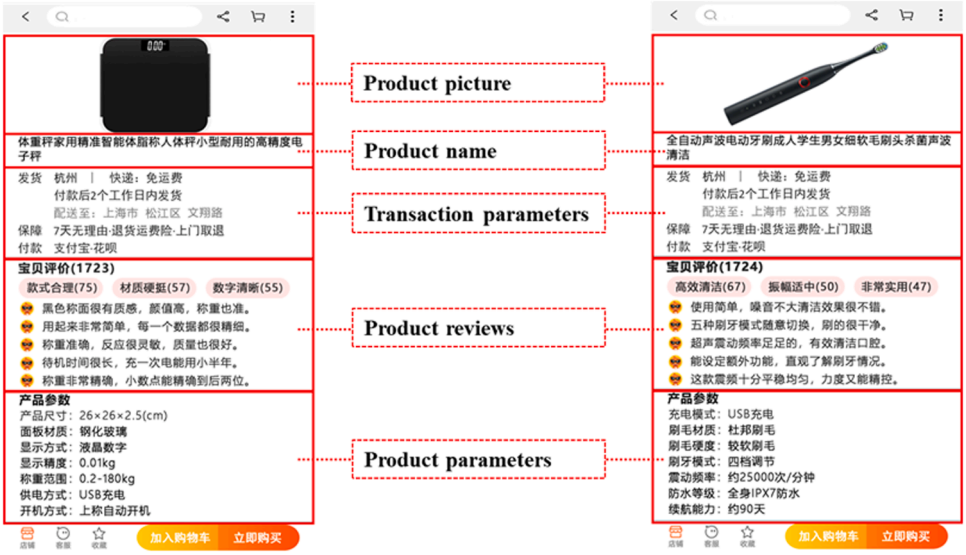


Fig. 7. Picture of experimental stimulus. Product pictures, product names, transaction parameters, product reviews, and product parameters are presented from top to bottom.

pressed "2" to proceed to the purchase intention rating page. A blank page was displayed for 600 ms before the purchase intention screen to help participants recover their attention and alleviate cognitive load between processing different stimuli [115]. During this process, the eye tracker automatically recorded the data of the participant's eye movements.

At the end of the experiment, participants completed a questionnaire based on the eye-tracking experiment. They rated their cognitive load using 2 items (Cronbach's $\alpha = 0.77$) [91], and perceived crowding using 5 items on a 7-point scale (Cronbach's $\alpha = 0.86$) [74], as detailed in Appendix A. Finally, the participants filled out demographic information, such as gender, and age. The entire experiment, which lasted approximately 20 min, is illustrated by the schematic diagram of the eye-tracking experiment shown in Fig. 8.

7.3. Measures

Fixation counts refer to the number of fixation points in a specific area. Fixation duration refers to the period taken to gaze at a specific area. Fixation counts and fixation durations directly indicate where consumers focus their visual attention and how intensely [4,15,71,86]. More fixation counts and longer fixation durations signify heightened attention [3] and sustained engagement in information processing [50, 99,104]. Attention refers to the actual cognitive resources an individual spends on the target [24]. The amount of attention is affected by focused attention [80], which determines both the location and depth of an individual's information processing [62]. Therefore, attention in turn reflects the degree of focused attention.

The percentage of fixation counts/durations in AOI to total fixation counts/duration. This metric serves to comprehend the degree of participants' attention toward specific areas, reflecting the information processing patterns [113]. A higher percentage indicates greater attention expended by participants on specific regions.

7.4. Data collection and analysis

The eye-tracking device used in this study was the EyeLink 1000 plus Desktop Mount system (http://srresearch.com) with a 1000-Hz sampling rate. It uses an infrared camera to capture video images of each participant's eye. Product information pages were presented by E-prime 3.0 (Psychology Software Tools, Pittsburgh, Pennsylvania, USA) on a 21-inch monitor with a resolution of 1080 px width × 1920 px height. During the experiment, participants rested their chins on a chin rest, sat comfortably, and kept their heads static still to improve the quality of eye-tracking data and ensure precise tracking of eye movements [11]. The distance between the monitor and the chin rest was approximately 85 cm. Eye-tracking data were recorded automatically after participants completed a 9-point calibration test.

The data were analyzed using the EyeLink Data Viewer software (SR Research Company, North America), and matched with the eye-tracker. In our experiment, the entire image, product picture, product name, transaction parameters, product reviews, and product parameters were selected as the areas of interest (AOIs), as exhibited in Appendix B. Statistical analysis was performed also using the SPSS (IBM SPSS 22.0, Chicago, IL) software package and the primary analysis method was repeated measures ANOVA analysis. The Greenhouse-Geisser correction was applied for the violation of the sphericity assumption in ANOVA where appropriate [39].

7.5. Results

7.5.1. Manipulation checks of social crowding

Participants in the high social crowding group reported stronger perceived crowding ($M_{high} = 4.73$, $SD = 1.06$) than those in the low social crowding group ($M_{low} = 3.69$, $SD = 1.31$; $t(103) = -4.462$, $p < 0.001$), which suggested the manipulation of social crowding was successful.

7.5.2. Behavioral results

Purchase intentions. A mixed design repeated measure analysis of variance (ANOVA) revealed that both the main effects of social crowding ($F(1, 103) = 4.960$, $p = 0.028$, $\eta_p^2 = 0.046$) and product type ($F(1, 103) = 5.679$, $p = 0.019$, $\eta_p^2 = 0.052$) were significant. The purchase intentions in high social crowding ($M_{high} = 4.14$, $SE = 0.10$) were lower than in low social crowding ($M_{low} = 4.46$, $SE = 0.10$), and H1 was supported. The purchase intentions for search products ($M_{search} = 4.16$, $SE = 0.11$) were lower than for experience products ($M_{experience} = 4.44$, $SE = 0.08$).

Moreover, the interaction effect between social crowding and product type was also significant ($F(1, 103) = 7.679$, $p = 0.007$, $\eta_p^2 = 0.069$). Further simple effect analysis indicated that the significant difference in purchase intentions for search products ($F(1, 103) = 9.336$, $p = 0.003$, $\eta_p^2 = 0.083$) between high ($M_{high\text{-}search} = 3.84$, $SE = 0.15$) and low social crowding ($M_{low\text{-}search} = 4.48$, $SE = 0.15$), supporting H4a. However, it was not significant for experience products ($F(1, 103) = 0.003$, $p = 0.955$, $\eta_p^2 < 0.001$), supporting H4b. The results are shown in Fig. 9.

7.5.3. Eye-tracking results

Heat map analysis. Before our quantitative analysis, we conducted a heat map analysis to qualitatively show consumers' visual attention to

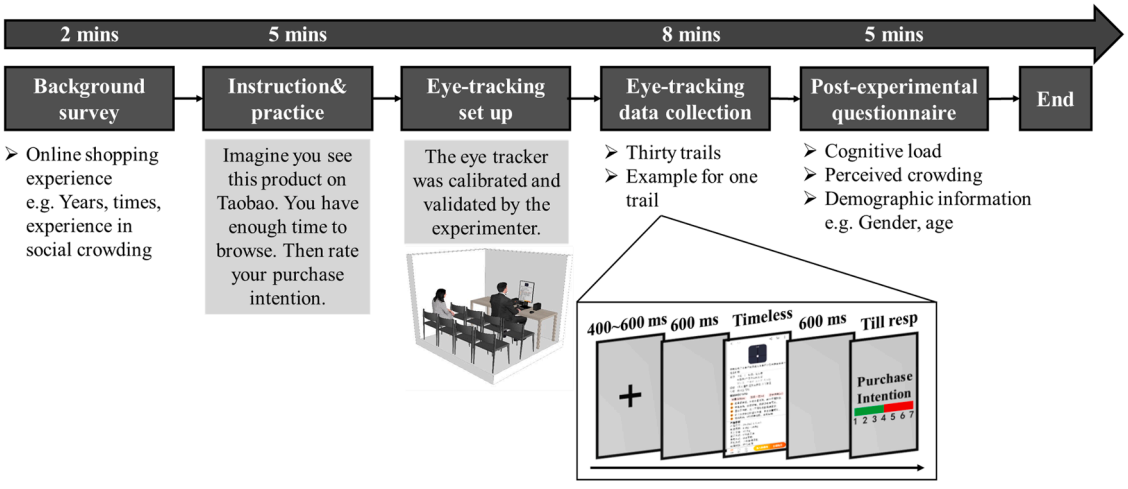


Fig. 8. Flow chart of the eye-tracking experiment.

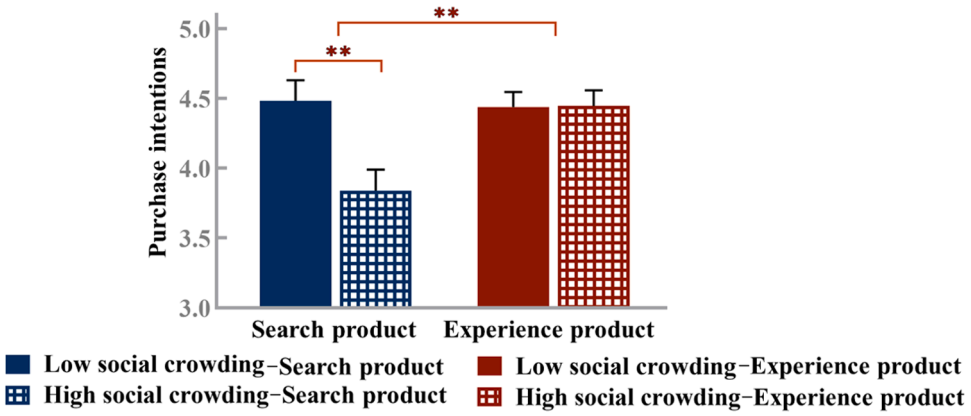


Fig. 9. Purchase intentions for search and experience products.
Error bars denote standard error across participants. $^{**}p < 0.01$.

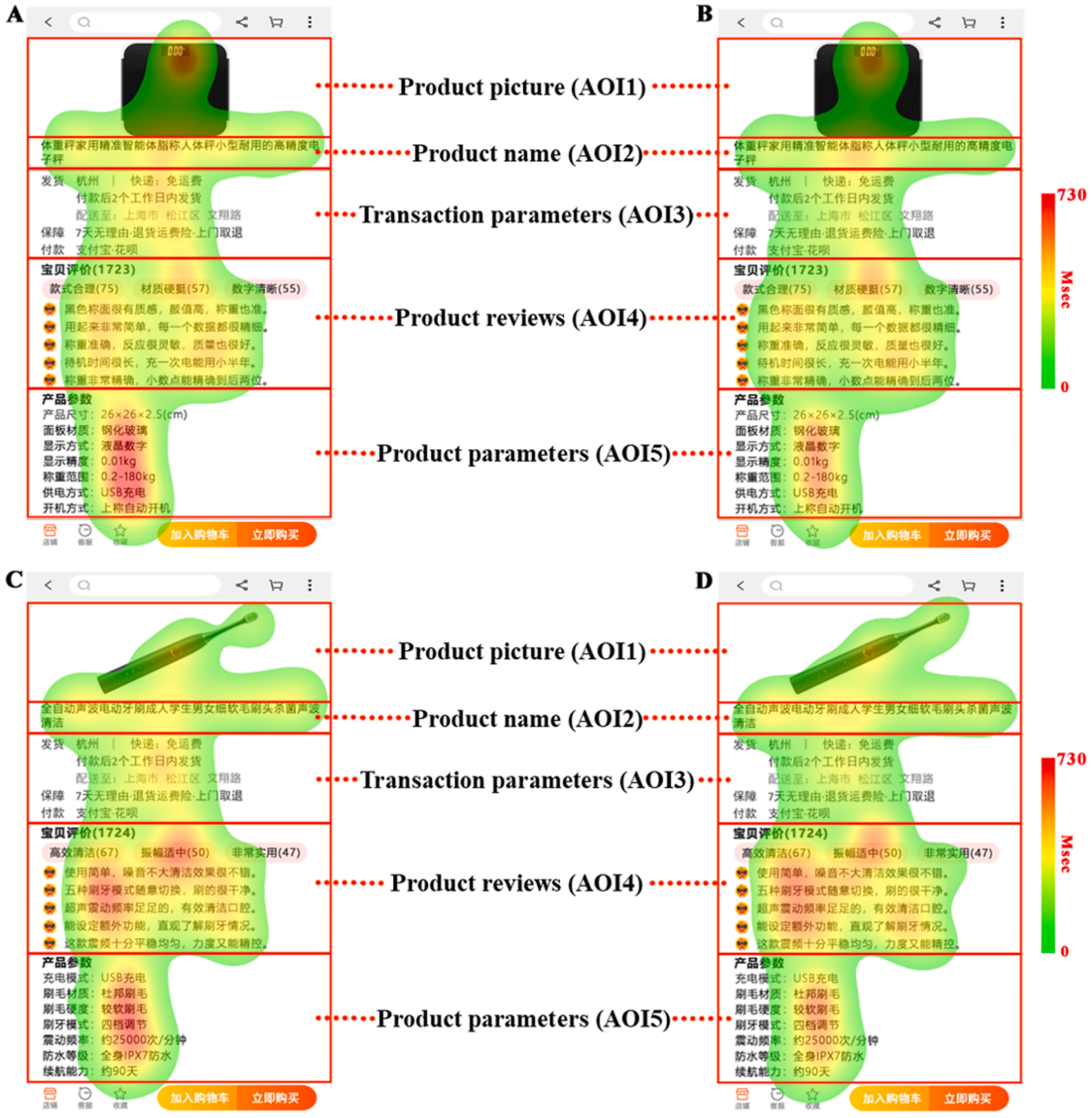


Fig. 10. Heat map of AOIs. **A**. Search product-low social crowding. **B**. Search product-high social crowding. **C**. Experience product-low social crowding. **D**. Experience product-high social crowding.

products, as shown in Fig. 10. The heat maps feature only one product image for each product type. For the search product, a weight scale, which best represents search indicators, was chosen. For the experience product, an electric toothbrush, embodying the most distinctive experience indicators, was selected. Heat maps can intuitively show the degree of individuals' visual attention and provide a more direct understanding of fixation results [105,117]. In the heat map, the shade of the color manifests the degree of consumers' fixations. Red indicates the highest level of fixation, followed by yellow and green, and areas without color attract no fixations.

According to Fig. 10A, for search products under low social crowding, the color of product parameters (AOI5) was redder compared to the other AOIs (AOI1/2/3/4), indicating individuals' heightened attention to product parameters (AOI5) during information processing. According to Fig. 10C, for experience products under low social crowding, the color of product reviews (AOI4) was redder compared to the other AOIs (AOI1/2/3/5), suggesting individuals' increased attention to AOI4 during information processing.

Comparing Fig. 10A & B, for search products under high social crowding, the colors of transaction parameters (AOI3) and product parameters (AOI5) were lighter than low social crowding, indicating reduced fixation on transaction parameters (AOI3) and product parameters (AOI5). However, product picture (AOI1), product name (AOI2), and product reviews (AOI4) showed no significant differences. Similarly, comparing Fig. 10C & D, for experience products under high crowding, the colors of transaction parameters (AOI3) and product parameters (AOI5) were also lighter than low social crowding, suggesting reduced fixation on transaction parameters (AOI3) and product parameters (AOI5). And product picture (AOI1), product name (AOI2), and product reviews (AOI4) also showed no significant differences.

Regarding the specific effects of the interaction between social crowding and product type on AOIs, we will further analyze AOIs.

Total fixation counts result. We conducted a mixed design repeated measure analysis of variance (ANOVA) to compare the effects of social crowding (high vs. low) and product type (search vs. experience) in total fixation counts of the entire page. The main effect of social crowding ($F(1, 103)=4.404$, $p = 0.038$, $\eta_p^2 = 0.041$) was significant. The total fixation counts in high social crowding ($M_{high} = 43.89$, $SE = 3.24$) were lower than in low social crowding ($M_{low} = 53.37$, $SE = 3.15$). The main effect of product type ($F(1, 103)=14.673$, $p < 0.001$, $\eta_p^2 = 0.125$) was significant. The total fixation counts of search products ($M_{search} = 46.29$, $SE = 2.25$) were lower than experience products ($M_{experience} = 50.97$, $SE = 2.43$). Moreover, there was also a significant interaction effect between social crowding and product type ($F(1, 103)=4.436$, $p = 0.038$, $\eta_p^2 = 0.041$). Further simple effect analysis indicated that the difference in total fixation counts of the entire page about search products between high ($M_{high\text{-}search} = 40.26$, $SE = 3.23$) and low social crowding ($M_{low\text{-}search} = 52.31$, $SE = 3.13$) was significant ($F(1, 103)=7.184$, $p = 0.009$, $\eta_p^2 = 0.065$), supporting H4a. But it was not significant for experience products ($F(1, 103)=2.021$, $p = 0.158$, $\eta_p^2 = 0.019$), supporting H4b. We summarized all the findings in Appendix A.

The mediation model test with total fixation counts. We adopted PROCESS model 4 [45] to conduct the mediation analysis. We found that total fixation counts fully mediated the effect of social crowding on purchase intentions (*indirect effect*=−0.111, SE = 0.044, 95 %CI=[−0.216, −0.040]), supporting H2.

The moderated mediation model test with product type as the moderating variable and total fixation counts as the mediating variable. We performed a moderated mediation analysis by using the PROCESS Model 7 [45]. While the index of moderated mediation did not reach significance (*indirect effect*=−0.206, SE = 0.130, 95 %CI=[−0.463, 0.051]), we examined the separate indirect effects at each product type based on our a priori predictions [21,46]. In other words, while we did not have predictions for what might drive purchase intentions for experience products (*indirect effect*=−0.049, SE = 0.039, 95 %CI=[−0.156, 0.007]), we did predict that for search products (*indirect effect*=−0.187, SE = 0.082, 95 %CI=[−0.387, −0.059]), lowered total fixation counts would explain the decreased purchase intentions. Overall, these results supported H3a, H3b, H4a, and H4b.

The percentage of fixation counts in AOI to total fixation counts results. A mixed design repeated measure analysis of variance (ANOVA) was conducted in the percentage of fixation counts in AOI to total fixation counts for each AOI, respectively. The results showed that there was no significant interaction effect between social crowding and product type for all AOI.

7.5.4. Supplementary analysis

We also conducted a mixed design repeated measure analysis of variance (ANOVA) on fixation counts for each AOI, with the results presented in Appendix C. Additionally, we performed a similar analysis to examine the effects of social crowding and product type on total fixation duration for the entire page, fixation duration for each AOI, and the percentage of fixation duration in AOI to total fixation duration for each AOI. These results on fixation durations were consistent with our earlier findings, further supporting the robustness of our results. We put these results in Appendix C.

7.6. Discussion

In Study 3, we validated the main effect of social crowding and its interaction effect with product type, showcasing the robustness of the behavioral phenomena. Notably, we identified that fixation counts on the overall page for search products were reduced under conditions of high social crowding compared to low social crowding, rather than experience products. Furthermore, we identified that total fixation counts played a mediating role in the impact of social crowding on consumers' mobile consumption behavior across different product types, explaining the effect of social crowding. This finding directly supports the notion that reduced focused attention impairs overall attention allocation and disrupts information processing, ultimately affecting purchase intentions.

In the AOI analysis, social crowding had no significant effect on product picture (AOI1), product name (AOI2), or product reviews (AOI4), but it did affect transaction parameters (AOI3) and product parameters (AOI5). This suggests that social crowding impedes the browsing of product parameters, which are crucial for evaluating search products, thereby reducing purchase intention for these products. This behavior may be explained by the fact that processing parameter information for search products requires sustained cognitive resources and focused attention, which are more difficult to maintain in crowded environments. As a result, participants may simplify their decision-making by selectively avoiding information that demands greater attentional effort [114]. However, since product parameters are less important for evaluating experience products [72], it does not influence their purchase intentions. The primary focus of this study lies in examining the influence of social crowding and the product itself, rather than delving into the transaction parameters of the purchasing process. As such, transaction parameters (AOI3) were not given undue attention.

8. General discussion

The current research employed a scenario-based questionnaire, field research, and an eye-tracking experiment to explore whether and how social crowding impacts consumers' mobile purchase behavior based on the environmental load theory. We find the negative impact of social crowding on mobile purchase intentions. Specifically, the purchase intentions for search products are lower in high social crowding compared to low social crowding. Furthermore, we identify a mediate and moderate model from the behavioral level, which suggested the mediating effect of focused attention and the moderating effect of product type. This effect cannot be found for the experience product. Regarding the eye-tracking results, we find that the total fixation counts on search

products are significantly lower in high social crowding than in low crowding. We further revealed that total fixation counts, serving as a proxy for focused attention, mediated the impact of social crowding on consumers' mobile purchase intentions across different product types. AOI analysis shows that the fixation counts of product parameters (AOI5) are lower in high social crowding than in low social crowding.

Theoretically, this research has the following contributions: First, our research promotes the development of social crowding theory by revealing the effect of social crowding on consumer behaviors in mobile shopping and underlying mechanisms. We identify a moderated mediation model in social crowding on mobile purchase intentions, with focused attention as a mediator and the product type as a moderator. Eye-tracking results indicated that total fixation counts, as an indicator of focused attention, functioned as a mediator in the relationship between social crowding and consumers' mobile purchase intentions across different product types. These findings provide further evidence for the application of the environmental load theory in social crowding environments for the first time and further emphasize the robust applicability of this theory.

Second, this research contributes to the literature on product type by enhancing the understanding of differences in consumer behavior toward search versus experience products. Much previous literature has delved into how consumers respond differently to these two product types in various shopping settings, such as facing different recommenders [127], cultural influences [120], and web page designs [132]. However, previous studies have largely overlooked the setting of mobile shopping, which has become an integral part of our daily lives. We add to this stream of research by disclosing how consumers behave differently toward search versus experience products in mobile shopping with the consideration of social crowding. Notably, we explore the cognitive mechanisms underlying the effect of social crowding (i.e., focused attention to products and information processing), thereby enhancing our understanding.

Third, this research advances the literature on information processing by offering novel insights into individuals' attention and information processing patterns in social crowding environments. Specifically, we identify that in low social crowding, where ample cognitive resources are available, individuals' attention is high, allowing for deep engagement with the information. Conversely, a high social crowding environment leads to a reduction in attention, resulting in the less thorough evaluation of complex information. Eye-tracking results further reveal that social crowding significantly reduces individuals' fixation counts on product parameters, indicating a tendency to superficially skim through complex information (product parameters) to mitigate the cognitive load induced by social crowding.

This study also has implications for practice. First, considering that social crowding negatively affects consumers' purchase intentions for search products, especially in processing product parameters (AOI5), optimizing the layout of the search product page becomes pivotal. For instance, integrating key parameter information into product images or prioritizing product reviews that contain parameter information can be effective strategies. Furthermore, optimizing the product-push system of mobile e-commerce is another avenue. For instance, during peak commuting hours, mobile e-commerce operators can strategically reduce promotions for search products and redirect efforts toward more engaging content that alleviates stress or entertains commuters.

Second, given that the negative impact of social crowding on purchase intentions primarily stems from reduced focused attention and impeded information processing, platform providers and interface designers can explore ways to reduce distractions and enhance consumers' focused attention. For example, during peak commuting hours, using larger fonts and minimizing on-screen text can help consumers process information more easily. In addition, incorporating videos, images, or animations can better capture attention and promote deeper focused attention to the content.

9. Conclusion

Given the widespread use of mobile devices and the escalating prevalence of social crowding, a deeper understanding of how social crowding affects mobile shopping is necessary for e-commerce platforms seeking to enhance efficiency and effectiveness. This research utilized a scenario-based questionnaire and field research to progressively reveal the moderating role of product type and the mediating role of focused attention in the relationship between social crowding and mobile purchasing behavior. Eye-tracking results further demonstrate that total fixation counts, used as an indicator of focused attention, mediated the impact of social crowding on consumers' mobile purchase intentions across different product types. This decline in focused attention predominantly impedes the information processing of search products, leading to a decrease in purchase intentions. These findings provide useful guidance for e-commerce platforms and marketing practices in optimizing recommendation systems and webpage design to better align with consumers' information processing, ultimately improving purchase conversion rates and user experience in social crowding environments.

10. Limitation

Although this study enhances our comprehension of the impact of social crowding and product types on purchase intentions, we recognize certain limitations in our research. Despite our endeavors to align experimental settings closely with real-life scenarios across all studies. The issue of ecological validity still exists. For example, participants were required to stabilize their heads by placing their chin on a chinrest during the eye-tracking experiments, which is an unnatural scenario. This might introduce a level of artificiality to certain behaviors. Future research could explore more naturalistic experimental designs to gain a comprehensive understanding of the relationship between consumer behavior and eye movements.

CRediT authorship contribution statement

Jia Jin: Writing – review & editing, Writing – original draft, Supervision, Methodology, Funding acquisition, Data curation, Conceptualization. **Qingsong Liu:** Writing – review & editing, Writing – original draft, Visualization, Validation, Formal analysis, Conceptualization. **Yufeng Yang:** Writing – review & editing, Software. **Baojun Ma:** Software, Data curation. **Yu Pan:** Supervision, Resources, Project administration.

Declaration of competing interest

All the authors declare that there are no conflicts of interest.

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Appendix A. Tables

Tables A1, A2, A3, A4

Table A1
Measurements scales.

Scales (1 = Strongly disagree - 7 = Strongly agree)	
Perceived crowding	
Q1	This environment seemed very spacious for me.
Q2	This environment was a little too constrained for me to do mobile shopping.
Q3	This environment was a little too noisy for me to do mobile shopping.
Q4	This environment seemed less flow of people.
Q5	This environment seemed like a huge amount of people.
Product type	
Q1	For this product, I can obtain a lot of product information through the web page before purchasing and evaluate whether it meets my needs through this information.
Q2	As for the product information of the product, I can get it through the website before purchasing and use the information to evaluate whether it meets my needs.
Q3	As for the product information of the product, I can get it mostly after the purchase and evaluate whether it meets my needs through this information.
Q4	For this product, I can only know whether it meets my needs after I buy or experience it.
Focused attention	
Q1	In this environment, I can be deeply engrossed in shopping for USB drives (sunglasses) on my phone.
Q2	In this environment, I can be absorbed intently in purchasing USB drives (sunglasses) on my phone.
Q3	In this environment, my attention is focused on buying USB drives (sunglasses) via mobile shopping.
Q4	In this environment, I concentrate fully on shopping for USB drives (sunglasses) using my phone.
Purchase intention	
Q1	I am likely to purchase the product.
Q2	I am inclined to buy the product.
Q3	It's possible for me to buy the product.
Cognitive load	
Q1	How much thought did you put into the product evaluation in this environment?
Q2	What do you think of the difficulty of the task just performed in the experiment?

Table A2
Descriptive statistical analysis of pretest stimulus pictures questionnaires.

Picture number	Min	Max	M	SD	Alpha coefficient
P2	4.40	7.00	6.600	0.451	0.880
P14	4.40	7.00	6.527	0.450	0.898
P10	4.00	7.00	6.315	0.630	0.877
P8	4.40	7.00	6.291	0.593	0.800
P16	4.20	7.00	6.285	0.905	0.792
P6	3.40	7.00	6.164	1.214	0.790
P12	4.20	7.00	6.164	0.626	0.821
P18	2.60	7.00	5.933	1.399	0.878
P4	3.00	7.00	5.412	1.260	0.912
P20	2.00	7.00	5.279	1.615	0.826
P9	1.00	4.80	2.297	1.243	0.812
P5	1.00	4.80	2.152	1.188	0.829
P11	1.00	5.20	2.109	1.713	0.856
P7	1.00	5.80	2.091	1.715	0.802
P17	1.00	4.60	1.897	1.438	0.791
P3	1.00	4.60	1.891	1.203	0.834
P19	1.00	5.20	1.873	1.587	0.804
P1	1.00	4.80	1.824	1.272	0.825
P13	1.00	4.60	1.782	1.106	0.835
P15	1.00	4.60	1.552	0.838	0.891

Table A3
Descriptive statistical analysis of pretest questionnaires for search and experience products.

	Search index		Experience index		t (39)	*p*
	M	*SD*	M	*SD*		
Weight scale	3.88	0.59	3.33	0.98	3.20	0.003
USB drive	3.86	0.64	3.28	0.73	3.28	0.002
Power strip	3.80	0.58	3.30	0.99	2.52	0.016
Mug	3.85	0.66	3.65	0.75	1.34	0.189
Mobile phone shell	3.82	0.68	3.78	0.92	0.26	0.800
Mobile phone data cable	3.76	0.69	3.60	0.93	0.92	0.366
Mouse	3.79	0.70	3.80	0.92	−0.07	0.946
Battledore	3.80	0.50	3.85	0.93	−0.31	0.756
Computer backpack	3.75	0.55	3.85	0.85	−0.6	0.546
Desk lamp	3.65	0.78	3.78	0.95	−0.63	0.531
Wallet	3.68	0.70	3.85	0.83	−0.95	0.349
Bluetooth headset	3.49	0.64	3.82	0.84	−2.17	0.036
Electric toothbrush	3.49	0.75	3.94	0.76	−2.64	0.012
Sunglasses	3.26	0.87	3.89	0.76	−3.82	0.000

Table A4
Summary of the studies.

	Variable	**Factor**	**F**	***df***	***p***	η_p^2	**Mean**
Study 1	Purchase intentions	Social crowding	164.287	427	< 0.001	0.278	High: 3.39 (0.08) Low: 4.85 (0.08)
		Product type	63.201	427	< 0.001	0.129	S: 3.67 (0.08) E:4.58 (0.08)
		Social crowding × Product type	136.352	427	< 0.001	0.242	High-S: 2.27 (0.13) Low-S: 5.07 (0.11)
Study 2	Purchase intentions	Social crowding	9.453	198	0.002	0.046	High: 4.00 (0.12) Low: 4.54 (0.12)
		Product type	4.923	198	0.028	0.024	S: 4.09 (0.12) E: 4.44 (0.12)
		Social crowding × Product type	9.259	198	0.003	0.045	High-S: 3.58 (0.17) Low-S: 4.60 (0.17)
Study 3	Purchase intentions	Social crowding	4.960	103	0.028	0.046	High: 4.14 (0.10) Low: 4.46 (0.10)
		Product type	5.679	103	0.019	0.052	S: 4.16 (0.11) E: 4.44 (0.08)
		Social crowding × Product type	7.679	103	0.007	0.069	High-S: 3.84 (0.15) Low-S: 4.48 (0.15)
	Total fixation counts	Social crowding	4.404	103	0.038	0.041	High: 43.89 (3.24) Low: 53.37 (3.15)
		Product type	14.673	103	< 0.001	0.125	S: 46.29 (2.25) E: 50.97 (2.43)
		Social crowding × Product type	4.436	103	0.038	0.041	High-S: 40.26 (3.23) Low-S: 52.31 (3.13)
	Product picture (AOI1)	Social crowding	2.620	103	0.109	0.025	N/A
		Product type	2.336	103	0.130	0.022	N/A
		Social crowding × Product type	3.152	103	0.079	0.030	N/A
	Product name (AOI2)	Social crowding	0.062	103	0.804	0.001	N/A
		Product type	2.653	103	0.106	0.025	N/A
		Social crowding × Product type	12.421	103	< 0.001	0.108	High-S: 6.69 (0.59) High-E: 7.60 (0.59)
	Transaction parameters (AOI3)	Social crowding	4.980	103	0.028	0.046	High: 5.18 (0.39) Low: 6.40 (0.38)
		Product type	7.746	103	0.006	0.070	S: 5.51 (0.28) E: 6.07 (0.30)
		Social crowding × Product type	1.418	103	0.237	0.014	N/A
	Product reviews (AOI4)	Social crowding	2.226	103	0.139	0.021	N/A
		Product type	11.233	103	0.001	0.098	S: 14.61 (0.89) E: 16.53 (0.99)
		Social crowding × Product type	2.264	103	0.136	0.022	N/A
	Product parameters (AOI5)	Social crowding	5.118	103	0.026	0.047	High: 8.82 (1.02) Low: 12.05 (1.00)
		Product type	15.401	103	< 0.001	0.130	S: 9.81 (0.69) E: 11.07 (0.77)
		Social crowding × Product type	0.362	103	0.548	0.004	N/A

Notes: S = Search products; E = Experience products.

Appendix B. Figures

Figs. B1, B2, B3, B4



A. High social crowding B. Low social crowding

Fig. B1. Social crowding stimuli materials for Study 1.



A. Weight scale (Search product) B. Electric toothbrush (Experience product)

C. USB drive (Search product) D. Sunglasses (Experience product)

E. Power strip (Search product) F. Bluetooth headset (Experience product)

Fig. B2. Product stimuli materials for Studies 1 & 2 & 3.



A. High social crowding scene



B. Low social crowding scene

Fig. B3. Photographs of the experimental scene in Study 2.



Fig. B4. Examples of experimental stimulus.

Appendix C. Supplementary analysis

Cognitive load. Independent-samples *t*-test showed that participants in the high social crowding group reported stronger cognitive load ($M_{high\ crowding} = 4.25$, $SD = 1.11$) than those in the low social crowding group ($M_{low\ crowding} = 3.79$, $SD = 0.96$; t (103)=−2.265, $p = 0.013$).

AOI analysis for fixation counts. We also conducted a mixed design repeated measure analysis of variance (ANOVA) to examine the effects of social crowding (high vs. low) and product type (search vs. experience) in the fixation counts for each AOI. For product picture (AOI1), neither the main effect of social crowding (F (1, 103)=2.620, $p = 0.109$, $\eta_p^2 = 0.025$) nor the main effect of product type (F (1, 103)=2.336, $p = 0.130$, $\eta_p^2 = 0.022$)

was significant. Additionally, the interaction effect between social crowding and product type was also not significant (F (1, 103)=3.152, $p = 0.079$, $\eta_p^2 = 0.030$).

For product name (AOI2), neither the main effect of social crowding (F (1, 103)=0.062, $p = 0.804$, $\eta_p^2 = 0.001$) nor the main effect of product type (F (1, 103)=2.653, $p = 0.106$, $\eta_p^2 = 0.025$) was significant. However, the interaction effect was significant (F (1, 103)=12.421, $p < 0.001$, $\eta_p^2 = 0.108$). Further simple effect analyses indicated that there were no significant differences between high and low social crowding, whether for search products (F (1, 103)=1.017, $p = 0.316$, $\eta_p^2 = 0.010$) or experience products (F (1, 103)=0.217, $p = 0.603$, $\eta_p^2 = 0.003$). The interaction effect was driven by a significant simple effect in high social crowding (F (1, 103)=12.909, $p < 0.001$, $\eta_p^2 = 0.111$), where the fixation counts for search products ($M_{\text{high crowding-search}} = 6.69$, SE = 0.59) were lower than those for experience products ($M_{\text{high crowding-experience}} = 7.60$, SE = 0.59).

For transaction parameters (AOI3), both the main effects of social crowding (F (1, 103)=4.980, $p = 0.028$, $\eta_p^2 = 0.046$) and the product type (F (1, 103)=7.746, $p = 0.006$, $\eta_p^2 = 0.070$) were significant. However, the interaction effect was not significant (F (1, 103)=1.418, $p = 0.237$, $\eta_p^2 = 0.014$).

For product reviews (AOI4), the main effect of social crowding was not significant (F (1, 103)=2.226, $p = 0.139$, $\eta_p^2 = 0.021$). But the main effect of product type was significant (F (1, 103)=11.233, $p = 0.001$, $\eta_p^2 = 0.098$). The fixation counts for AOI4 of search products ($M_{\text{search products}} = 14.61$, SE = 0.89) were lower than experience products ($M_{\text{experience products}} = 16.53$, SE = 0.99). However, the interaction effect was not significant (F (1, 103)= 2.264, $p = 0.136$, $\eta_p^2 = 0.022$).

For product parameters (AOI5), both the main effects of social crowding (F (1, 103)=5.118, $p = 0.026$, $\eta_p^2 = 0.047$) and the product type (F (1, 103)=15.401, $p < 0.001$, $\eta_p^2 = 0.130$) were significant. The fixation counts for AOI5 in high social crowding ($M_{\text{high crowding}} = 8.82$, SE = 1.02) were lower than in low social crowding ($M_{\text{low crowding}} = 12.05$, SE = 1.00). The fixation counts for AOI5 of search products ($M_{\text{search products}} = 9.81$, SE = 0.69) were lower than experience products ($M_{\text{experience products}} = 11.07$, SE = 0.77). However, the interaction effect was not significant (F (1, 103)=0.362, $p = 0.548$, $\eta_p^2 = 0.004$).

Total fixation durations result. We conducted a mixed design repeated measure analysis of variance (ANOVA) to compare the effects of social crowding (high vs. low) and product type (search vs. experience) in the total fixation durations of the entire page. The main effect of social crowding (F (1, 103)=4.23, $p = 0.04$, $\eta_p^2 = 0.04$) was significant. The total fixation durations in high social crowding ($M_{\text{high crowding}} = 9353.21$, SE = 693.51) were lower than in low social crowding ($M_{\text{low crowding}} = 11{,}342.65$, SE = 673.97). The main effect of product type (F (1, 103)=18.88, $p < 0.001$, $\eta_p^2 = 0.16$) was significant. The total fixation durations of search products ($M_{\text{search products}} = 9760.56$, SE = 477.49) were lower than experience products ($M_{\text{experience products}} = 10{,}935.30$, SE = 525.48). Moreover, there was also a significant interaction effect between social crowding and product type (F (1, 103)=4.80, $p = 0.03$, $\eta_p^2 = 0.05$). Further simple effect analysis indicated that the difference in total fixation durations of the entire page about search products between high ($M_{\text{high crowding}} = 8469.72$, SE = 684.85) and low ($M_{\text{low crowding}} = 11{,}051.41$, SE = 665.56) social crowding was significant (F (1, 103)=7.31, $p < 0.01$, $\eta_p^2 = 0.07$), supporting H4a. But it was not significant for experience products (F (1, 103)=1.77, $p = 0.19$, $\eta_p^2 = 0.02$), supporting H4b.

AOI analysis for fixation duration. We also conducted a mixed design repeated measure analysis of variance (ANOVA) to compare the effects of social crowding and product type in the fixation duration for each AOI, respectively. For product picture (AOI1), neither the main effect of social crowding (F (1, 103)=2.664, $p = 0.106$, $\eta_p^2 = 0.025$) nor the main effect of product type (F (1, 103)=3.034, $p = 0.085$, $\eta_p^2 = 0.029$) was significant. What's more, the interaction effect between social crowding and product type was also not significant (F (1, 103)=3.163, $p = 0.078$, $\eta_p^2 = 0.030$).

For product name (AOI2), neither the main effect of social crowding (F (1, 103)=0.155, $p = 0.695$, $\eta_p^2 = 0.002$) nor the main effect of product type (F (1, 103)=2.337, $p = 0.129$, $\eta_p^2 = 0.022$) was significant. However, the interaction effect was significant (F (1, 103)=9.375, $p = 0.003$, $\eta_p^2 = 0.083$). Further simple effect analyses indicated that there were no significant differences between high and low social crowding, whether for search (F (1, 103)=1.588, $p = 0.211$, $\eta_p^2 = 0.015$) or experience product (F (1, 103)=0.238, $p = 0.627$, $\eta_p^2 = 0.002$). The interaction effect was because the simple effect for high social crowding was significant (F (1, 103)=10.243, $p = 0.002$, $\eta_p^2 = 0.090$). That is, the fixation duration for search products ($M_{\text{search products}} = 1339.68$, SE = 125.21) is lower than for experience products ($M_{\text{experience products}} = 1569.67$, SE = 127.70).

For transaction parameters (AOI3), either the main effect of social crowding (F (1, 103)=4.038, $p = 0.047$, $\eta_p^2 = 0.038$) or the main effect of product type (F (1, 103)=4.202, $p = 0.043$, $\eta_p^2 = 0.039$) was significant. However, the interaction effect was not significant (F (1, 103)=2.577, $p = 0.111$, $\eta_p^2 = 0.024$).

For product reviews (AOI4), the main effect of social crowding was not significant (F (1, 103)=1.762, $p = 0.187$, $\eta_p^2 = 0.017$). However, the main effect of product type was significant (F (1, 103)=12.045, $p < 0.001$, $\eta_p^2 = 0.105$). The fixation duration for AOI4 of search products ($M_{\text{search products}} = 3022.53$, SE =183.93) was lower than experience products ($M_{\text{experience products}} = 3442.49$, SE = 201.03). However, the interaction effect was not significant (F (1, 103)=1.971, $p = 0.163$, $\eta_p^2 = 0.019$).

For product parameters (AOI5), either the main effect of social crowding (F (1, 103)=4.121, $p = 0.045$, $\eta_p^2 = 0.038$) or the main effect of product type (F (1, 103)=29.412, $p < 0.001$, $\eta_p^2 = 0.222$) was significant. The fixation duration for AOI5 in high social crowding ($M_{\text{high crowding}} = 2050.78$, SE = 236.96) was lower than in low social crowding ($M_{\text{low crowding}} = 2721.57$, SE = 230.29). The fixation duration for AOI5 of search products ($M_{\text{search products}} = 2191.24$, SE = 158.24) was lower than experience products ($M_{\text{experience products}} = 2581.10$, SE = 179.27). However, the interaction effect was not significant (F (1, 103)=0.010, $p = 0.919$, $\eta_p^2 = 0.000$).

The percentage of fixation duration in AOI to total fixation duration results. A mixed design repeated measure analysis of variance (ANOVA) was conducted in the percentage of fixation duration in AOI to total fixation duration for each AOI, respectively. The results showed that there was no significant interaction effect between social crowding and product type for all AOI.

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